György Buzsáki's Concept of "Action" in *The Brain from Inside Out*

In his book ***The Brain from Inside Out***, neuroscientist György Buzsáki completely flips the traditional view of neuroscience, defining **action** (or motor output) as the **primary and fundamental engine of brain function**.

For Buzsáki, action is not a passive response to an external stimulus, but holds three key roles:

- **The calibrator of meaning (Grounding):** The brain is not a blank slate that passively absorbs information from the world. At birth, it is full of spontaneous electrical activity patterns that are "pure noise" or words without meaning. It is only through **action in the world** and the examination of its consequences that the brain "calibrates" these intrinsic patterns, giving them meaning. Touching or moving an object is what allows the senses to interpret reality (for example, understanding that a stick in water looks broken but actually isn't).
- **The prerequisite for perception:** Traditionally, the pathway is thought to be *Perception* → *Decision* → *Action*. Buzsáki argues the exact opposite (the *Inside-Out* framework): evolution developed motor systems for action and survival first; sensors (vision, hearing, etc.) emerged only later to **refine and guide action itself**. Without the ability to act, pure perception would have no evolutionary utility.
- **The origin of cognition (Internalized action):** This is one of the most fascinating points. Buzsáki argues that higher cognitive processes (thought, memory, planning) are nothing more than **internalized actions that have been disconnected from the muscles**. When we plan for the future or remember the past, brain circuits use the exact same neural mechanisms as physical navigation in space, but simulate the action "offline", without actually moving the body.

In summary, for Buzsáki, action is **the tool with which the brain actively explores the world to test hypotheses** and make useful predictions for survival.

Bibliography

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